

Nancy Sotero Silva

PHD STUDENT · UNIVERSITÄT BIELEFELD

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“The mind that opens to a new idea never returns to its original size”

Summary

Born in 1995 (Caruaru, Brazil). Secondary education was completed in 2011. Following initial studies in Nursing (2012–2014), a Bachelor's in Speech, Language, and Hearing Sciences was obtained (2018), followed by a Residency in Rehabilitation (2021) and a Master's in Neuroengineering (2023). Since 2023, a Doctorate in Biology has been pursued at Bielefeld, Germany. I am currently working on the integration between the oculomotor and auditory systems, but I also maintain a keen interest in cognition, communication and hearing disorders and rehabilitation, and musical perception/performance.

Education

Bielefeld University

Bielefeld, Germany

PHD IN BIOLOGY

July 2023 - Currently

- Dissertation: Understanding audio-visual integration from the perspective of eye movement-related eardrums oscillations (EMREOs). Supervisor: Prof. Dr. Christoph Kayser

Santos Dumont Institute

Macaíba, Brazil

MSC IN NEUROENGINEERING; RESIDENCY ON PERSON WITH DISABILITIES HEALTH CARE

March 2019 - February 2023

- Master's thesis: Influence of stimulus complexity on cortical electrophysiological responses in adults with normal hearing through multichannel acquisition. Supervisor: Prof. Dr. Edgard Morya

Federal University of Paráíba

João Pessoa, Brazil

BSC IN SPEECH, LANGUAGE AND HEARING SCIENCES

April 2014 - July 2018

- Bachelor's thesis: “Absolute Pitch” in music graduation students: is there a relationship with temporal auditory processing? Supervisor: Prof. Dr. Hannalice Gottschalck Cavalcanti and Dr. Raphael Bender Chagas Leite

Research Expeince

Cognitive Neuroscience Lab, Bielefeld University

Bielefeld, Germany

SCIENTIFIC WORKER/DOCTORAL RESEARCH PROJECT

July 2023 - Currently

- Conducts in-ear recordings (using BioSemi EEG system), eye tracking (EyeLink), hearing tests and psychophysics experiments (PsychToolbox); as long as data curation, processing (FieldTrip), analysis and reporting.

Medical Image Processing Lab, EPFL

Geneva, Switzerland

MASTER'S RESEARCH INTERNSHIP

July - August 2022

- 2 months internship - Summer Research Program.
- Explored processing pipeline optimization (MATLAB) for fNIRS data (Artinis) elicited by cognitive and auditory stimuli. Inspected data from open databases, tested fNIRS processing tools, prepared presentation and written reports.

Santos Dumont Institute

Macaíba, Brazil

MASTER'S AND RESIDENCY RESEARCH PROJECT

2019 - 2023

- Project: Electrophysiological, vascular/hemodynamic and pupillometric measures in cortical plasticity monitoring of adult hearing aids users. Data collection patients from a hearing rehabilitation service aiming the inclusion of follow-up assessments.
- Performed experiment design, data collection and analysis with electroencephalography (BrainVision), ABR, FFR, ERP, pupillometry (Mangold) and fNIRS (Nirx), clinical assessment and follow-up of patients (auditory assessment, hearing aids selection and fitting).

Federal University of Paráíba

João Pessoa, Brazil

BACHELOR'S RESEARCH PROJECT

2017 - 2018

- Identified the prevalence of absolute pitch among music undergraduates and its relation to auditory temporal resolution assessment tasks.
- Performed psychophysical experimentation, central auditory processing assessment and music cognition tests; as well as written reports and presentations.

OTHER RESEARCH AND COMMUNITY OUTREACH EXPERIENCES

2015 - 2021

- Communication and hearing assessment and rehabilitation for low-income patients, planning and execution of educational events for the community, organisation of scientific events, reports and presentations.
 - Transdisciplinary assessment of patients complaining of tinnitus
 - Speech therapy for patients with fluency disorders
 - Impact of noise on workers' health

Skills

Research, Data acquisition and analysis: Electrophysiology (EEG, ERP, ABR, FFR), fNIRS, Eyetracking (Pupillometry, Saccades). Programming: Python (basic), MATLAB (intermediate), LaTeX. Scientific Writing.

Languages, English (C1), Portuguese (native), Spanish (B2), German (A2). Beginner in French, Italian and Brazilian Sign Language.

Publication list

1. Sotero Silva, N., & Kayser, C. (2026). **Stability of Eye Movement-Related Eardrum Oscillations to acoustic and gravitational manipulations**. bioRxiv, 2026.04.16.718961. <https://doi.org/10.64898/2026.04.16.718961>
 2. Sotero Silva, N., Bröhl, F. & Kayser, C. (2026) **Sound lateralization ability is affected by saccade direction but not EMREOs**. Journal of Neurophysiology, 135(5), <https://doi.org/10.1152/jn.00555.2025>
 3. Wilroth, J., Sotero Silva, N., Tafakkor, A., Mesquita, B., Lau, B., Ip, E., Hannah, J. & Di Liberto, G. (2026). **fNIRS for continuous speech: considerations for hyperscanning**. bioRxiv (preprint), <https://doi.org/10.64898/2026.03.20.713212>
 4. Sotero Silva, N., Kayser, C. & Bröhl, F. (2025). **Unraveling eye movement-related eardrum oscillations (EMREOs): how saccade direction and tympanometric measurements relate to their amplitude and time course**. Hearing Research, 467. <https://doi.org/10.1016/j.heares.2025.109276>
 5. Cavalcanti, H.G., Lucena, M.H.M.S.L., Morya, E., Sotero Silva, N. & Paiva S.F. (2024) **Interface between Music and Audiology** (Chapter). In: Delgado, I.C., Andrade, K.C.L. & Rosa, M.R.D. (ed.). Speech Therapy Intervention in Hearing and Language. 1ed. Rio de Janeiro: Thieme Revinter, p. 56-66. ISBN: 978-65-5572-283-3 [available in portuguese]
 6. Sotero Silva, N., Evangelista, C.K.S., Cavalcanti, P.F., Balen, S.A., Cavalcanti, H.G. & Morya, E. (2022). **Review of Audiological Findings in SARS-CoV-2 Infection**. Journal of Neurological Disorders, 10(8). <https://doi.org/10.4172/2329-6895.10.9.514>
- Between 2020-2022: three other papers and one book chapter [available in portuguese].

MANUSCRIPTS IN PREPARATION

1. Sotero Silva, N., Vasconcelos, I.C., Alves, E.R.A., Monteiro, T.A., Spinelli, B.G., Miranda, D.C.S., Canto-Pereira, V.R., Balen, S.A. & Morya, E. (2026) **Auditory evoked potentials: does stimulus complexity influence responses?** In preparation
2. Campos, L.G.N., Sotero Silva, N., Alves, E.R.A., Silva, A.R.X., Soares, A.K.A., Dias, R.C.T., Brasil, F.L., Morya, E. & Rosa MRD. (2026) **Transcranial direct current stimulation reduces tinnitus discomfort**. In preparation.
3. Alexandre, V.G.A., Sotero Silva, N. & Vasconcelos, L.D.S. (2026) **Neuropsychology in speech therapy** (Chapter). In: Pinto, P.V.N., Vasconcelos, L.D.S. & Silva, F.T.M. (ed.) History of speech therapy in Brazil. 1. ed. Carapicuíba: Pró-fono. In press. [available in portuguese]

Presentations - Selected

ORAL PRESENTATIONS

2026	24th IMRF - International Multisensory Research Forum , Exploring the role of auditory efferences in Eye Movement-Related Eardrum Oscillations [Accepted for the Young Researcher Talk Session]	Genoa, Italy
2025	47th ECVP - European Conference on Visual Perception , The EMREOs time course differs between ipsi- and contralateral saccades	Mainz, Germany
2024	5th Workshop on Cognitive Neuroscience of Auditory and Cross-modal Perception , Comparing EMREOs induced by different sensory targets	Košice, Slovakia
2023	9th Neuroengineering Symposium , Comparing different stimulation complexities on LLAEPs for temporal and central cortical areas in adults	Macaíba, Brazil
2023	SIG Re(h)abilita - Universities Telemedicine Network , Speech Therapy and Neuroengineering	Virtual
2022	3rd VCCA - Virtual Conference on Computational Audiology , Electrophysiological and pupillometric measures in monitoring auditory cortex plasticity and listening effort post hearing aids	Virtual
2022	Rehabilitation Colloquium, McGill University , SARS-CoV-2 infection and audiological findings	Virtual

POSTERS

2026	Vision Augmented Hearing, Royal Society , EMREOs during noise and head-tilts [presenter]	Manchester, UK
2025	8th ICAC - International Conference on Auditory Cortex , Relation of EMREOs amplitude and time course with saccade direction and tympanometric measurements [presenter]	Maastricht, The Netherlands
2025	10th ISAAR - International Symposium on Auditory and Audiological Research , EMREOs repercussions on auditory spatial localization [presenter]	Nyborg, Denmark
2024	ESSEM - European Summer School on Eye Movements , EMREOs induced by auditory-guided saccades in different modalities of sound presentation [presenter]	Athens, Greece
2024	FENS Forum , EMREOs are induced by both visual- and auditory-guided saccades [presenter]	Vienna, Austria
2023	38th EIA - International Audiology Meeting , Pupillometric measurements and their relationship to auditory effort & Cognitive evoked potential in normal-hearing adults: active and passive oddball paradigms	Florianópolis, Brazil
2022	8th Neuroengineering Symposium , fNIRS data elicited by motor or auditory stimuli: Processing pipeline and tools for analysis feature extraction [presenter]	Macaíba, Brazil
2017	15th SLAN - Latin American Society of Neuropsychology , Musical perception in cochlear implant users: literature review [presenter] & Anatomy of the musical brain: Absolute Pitch [presenter]	Natal, Brazil

- Between 2013-2023: around five other oral presentations and 25 posters.

Awards, Honors and Stipends

2025	CNSP Hackathon , Awarded 2nd best participant	Maastricht, The Netherlands
2025	10th ISAAR, William Demant Foundation , Stipend for attendance and presentation	Nyborg, Denmark
2024	FENS, IBRO-PERC - Early Career Program Award , Stipend for a 3-weeks internship	Budapest, Hungary
2024	EPFL - Summer Research Program , Stipend for a 2-months internship	Geneva/Lausanne, Switzerland
2022	8th Neuroengineering Symposium, Santos Dumont Institute , Awarded one of the five best presentations	Macaíba, Brazil
2022	3rd VCCA , Awarded "Video Pitch Award"	Virtual
2019-2021	Ministry of Health, Federal Government of Brazil , Postgraduate Residency Stipend (24 months)	Natal, Brazil
2018	Federal University of Paraíba , Awarded "Academic Laurea" (graduation with honours) in BSc	João Pessoa, Brazil
2016-2018	Ministry of Education, Federal Government of Brazil , Stipends for teaching assistantship (12 months) and extracurricular internship (12 months) during BSc	João Pessoa, Brazil

Teaching and Supervision

2021-2022	Temporary Lecturer - UFRN , Otoneurology, Balance Disorders and Occupational Audiology courses and supervision in Hearing Rehabilitation internships	Natal, Brazil
2024-2026	Tutor/Lectures - Bielefeld University , Multisensory Perception and Perception and Action modules	Bielefeld, Germany

Music practice and instruction

2018-2020	BSc - interrupted , Degree in music - habilitation in classical singing	João Pessoa, Brazil
2018-2019	Research , Higher education in music: analysis of music undergraduation programs in Brazilian Northeast	João Pessoa, Brazil
2016-2018	Performance , Gazzi de Sá University Choir [choir and soloist]	João Pessoa, Brazil
2017-2018	Performance , Presto Ópera - classical singing for the community	João Pessoa, Brazil
2014-2015	Performance and community outreach , Speech Therapy Students' Choir	João Pessoa, Brazil